IGCSE 0607 Practice Questions — E3.2 — Combined Questions & Markscheme

Sketching graphs on a calculator

Total Questions: 6 — **Total Marks:** 45 Marks

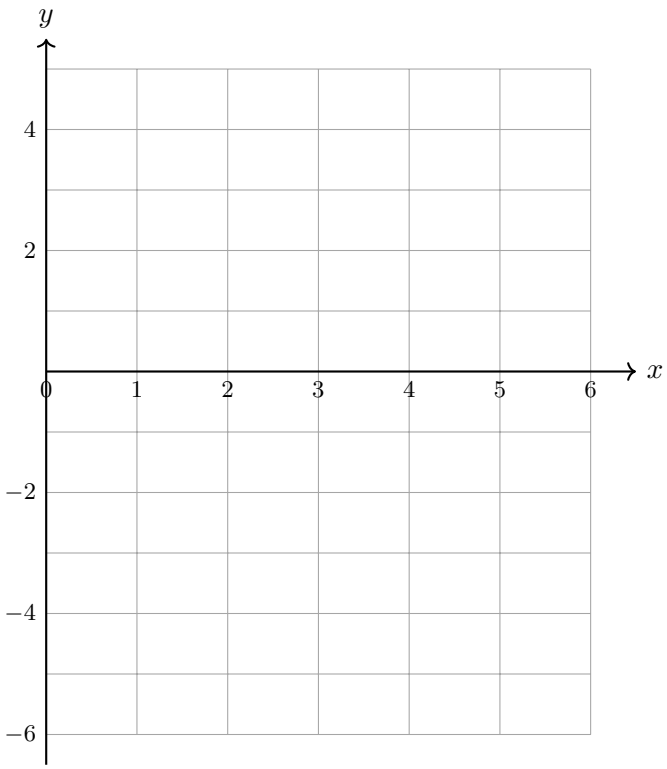
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- **E3.2** Sketching graphs on a calculator
  → **Paper 4 (Calculator)** *(6 Qs)*

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1 Let $f(x) = -x^2 + 6x - 5$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $0 \leq x \leq 6$.



[2]

(b) Find the zeros of $f(x)$.

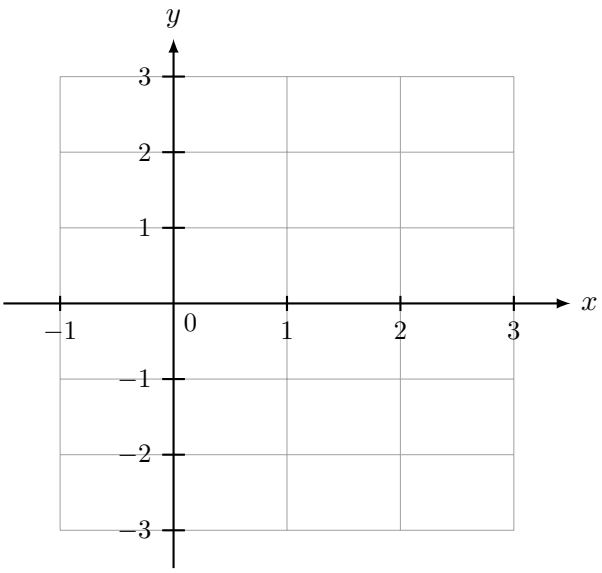
$x =$ ..... , $x =$ ..... **[2]**

(c) Find the coordinates of the vertex of the graph.

(..... , ...) **[2]**

2 Consider the function $f(x) = x^3 - 3x^2 + 2$ for $-1 \leq x \leq 3$.

(a) On the grid, sketch the graph of $y = f(x)$.



[2]

(b) Find the zeros of $f(x)$.

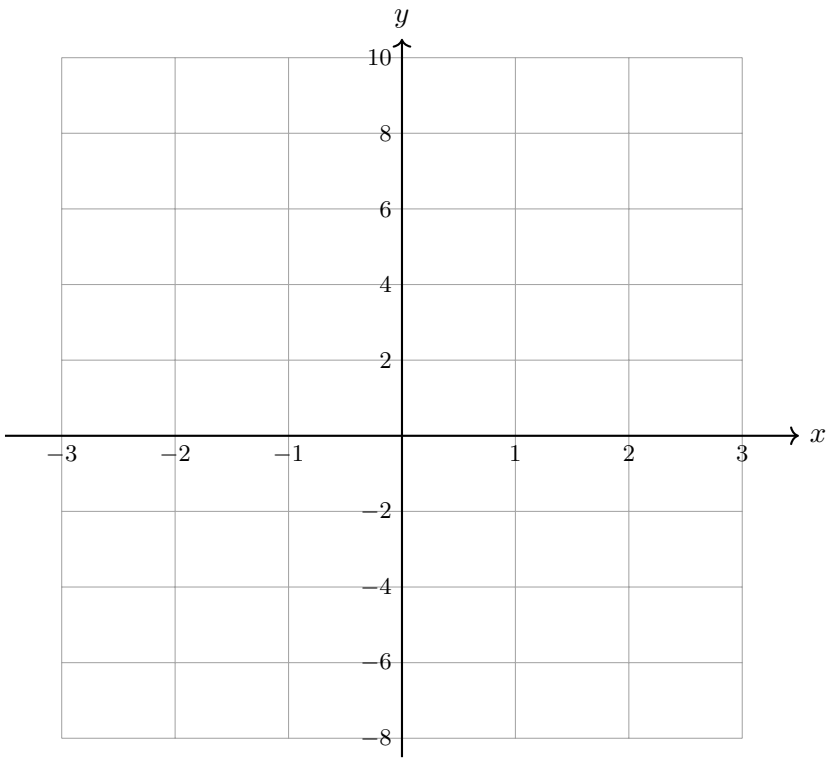
$x =$ ..... **[2]**

(c) Find the coordinates of the local minimum point of the graph.

(..... , ...) **[2]**

3 Let $f(x) = x^3 - 3x + 1$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $-2.5 \leq x \leq 2.5$.



[2]

(b) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

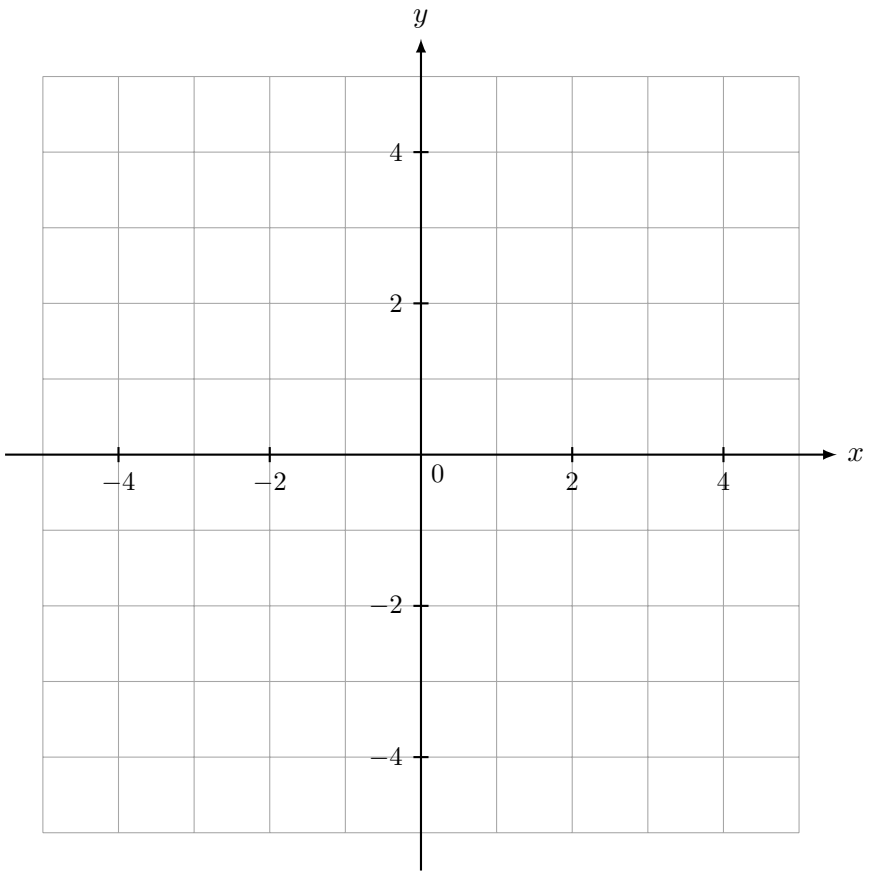
(c) On the same axes, sketch the graph of $y = 2x - 1$. **[1]**

(d) Find the x-coordinates of the points of intersection of the two graphs.

$x =$ ..... , $x =$ ..... , $x =$ ..... **[2]**

4 Let $f(x) = \dfrac{2}{x+1}$ and $g(x) = x - 2$ for $-5 \leq x \leq 5$.

(a) On the grid, sketch the graphs of $y = f(x)$ and $y = g(x)$.



[3]

(b) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

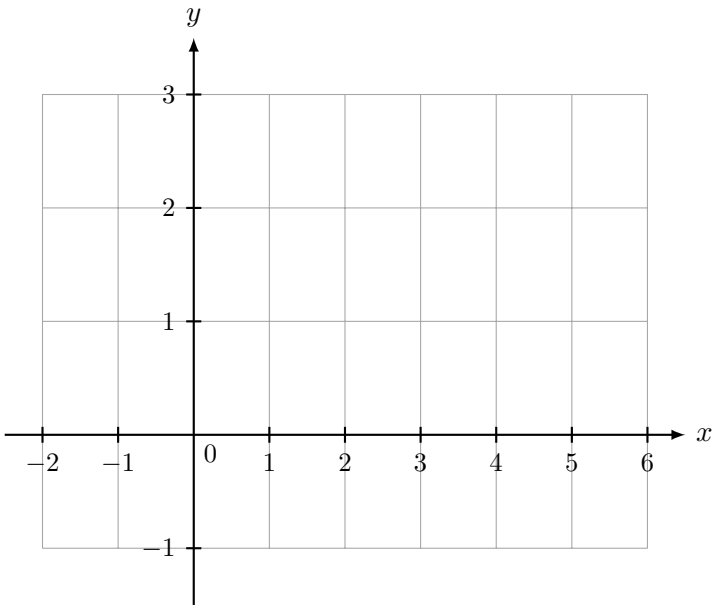
..... **[1]**

(c) Find the coordinates of the points of intersection of the two graphs.

Point 1 = ... , Point 2 = ... **[3]**

5 The function $f(x)$ is defined as $f(x) = \dfrac{x^2}{2^x}$ for the domain $-1 \leq x \leq 5$.

(a) On the grid, sketch the graph of $y = f(x)$.



[3]

(b) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

(c) Find the x-coordinates of the points where $f(x) = 0.5$.

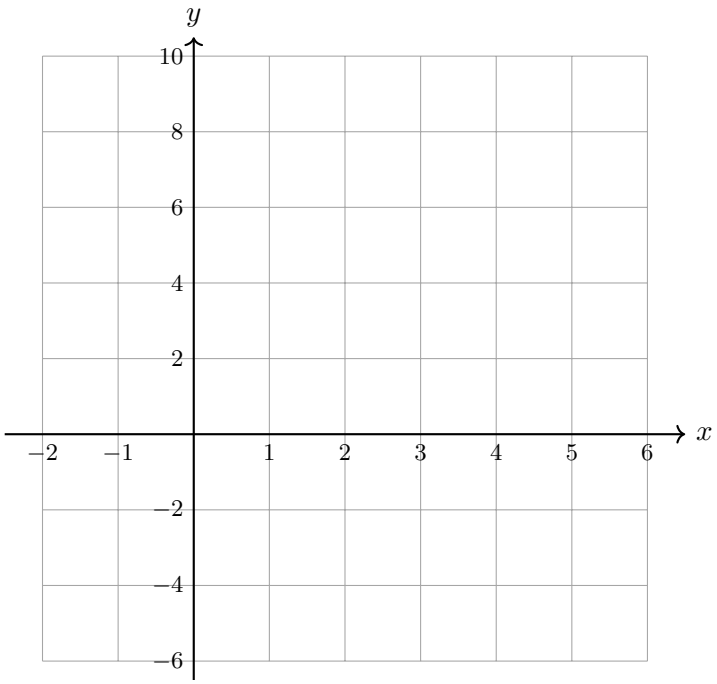
$x =$ ..... **[2]**

(d) Write down the range of values of k for which the equation $f(x) = k$ has exactly 3 different solutions.

..... **[2]**

6 Let $f(x) = \dfrac{x^2 - 2x + 4}{x - 2}$.

(a) On the axes provided, sketch the graph of $y = f(x)$ for $-2 \leq x \leq 6, x \neq 2$.



[3]

(b) Write down the equation of the vertical asymptote.

..... **[1]**

(c) Find the coordinates of the local maximum point.

(..... , ...) **[2]**

(d) Find the coordinates of the local minimum point.

(..... , ...) **[2]**

(e) The equation $f(x) = k$ has no real solutions.
Write down the range of values of k.

..... **[2]**

1 **(a)** Correct inverted U-shape parabola **[B1]**

Maximum approximately at $(3, 4)$ and crosses the x-axis at 1 and 5 **[B1]**

(b) $x = 1$ **[B1]**

$x = 5$ **[B1]**

(c) $(3, 4)$ **[B2]**

If B0, M1 for $x = 3$ or $y = 4$ identified

[Back to Question] *[Total Marks: 6]*

2 **(a)** Correct basic cubic shape with a local maximum and local minimum **[B1]**

Graph crosses y-axis at 2, has a maximum at $(0, 2)$ and a minimum at $(2, -2)$ **[B1]**

(b) Uses GDC to find roots **[M1]**

$-0.732, 1, 2.73$ (accept to 3 s.f.) **[A1]**

(c) Identifies correct turning point from GDC **[M1]**

$(2, -2)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

3 **(a)** Correct cubic shape with a local maximum and a local minimum **[B1]**

Local maximum approximately at $(-1, 3)$ and local minimum approximately at $(1, -1)$ **[B1]**

(b) $(-1, 3)$ **[B2]**

If B0, M1 for $x = -1$ or $y = 3$

(c) Correct straight line passing through $(0, -1)$ and $(2, 3)$ **[B1]**

(d) $x = 2$ **[B1]**

$x = 0.414$ and $x = -2.41$ **[B1]**

[Back to Question] *[Total Marks: 7]*

4 **(a)** Straight line for $g(x)$ passing through $(0, -2)$ and $(2, 0)$ **[B1]**

Two correct hyperbolic branches for $f(x)$ **[B1]**

Correct asymptotic behaviour near $x = -1$ **[B1]**

(b) $x = -1$ **[B1]**

(c) Sets up $\dfrac{2}{x+1} = x - 2$ or uses intersection feature on GDC **[M1]**

One correct coordinate pair: $(-1.56, -3.56)$ or $(2.56, 0.562)$ (to 3 s.f.) **[A1]**

Second correct coordinate pair **[A1]**

[Back to Question] *[Total Marks: 7]*

5 **(a)** Graph touches the origin $(0, 0)$ **[B1]**

Graph reaches a maximum in the first quadrant and then approaches the x-axis asymptotically **[B1]**

Curve sketched with reasonable accuracy from $x = -1$ to $x = 5$, with $f(-1) = 2$ **[B1]**

(b) Uses maximum feature on GDC **[M1]**

$(2.89, 1.13)$ (accept $x = 2.88$ to 2.89 and $y = 1.12$ to 1.13) **[A1]**

(c) Draws line $y = 0.5$ on GDC or sets up equation **[M1]**

$-0.589, 0.835, 5.38$ (all three required, accept to 3 s.f.) **[A1]**

(d) Recognises the horizontal line $y = k$ must lie between the local minimum (0) and local maximum (1.13) **[M1]**

$0 < k < 1.13$ **[A1]**

[Back to Question] *[Total Marks: 9]*

6 **(a)** Correct shape with two distinct branches **[B1]**

Left branch passing through $(0, -2)$ and tending downwards as x

2^- [B1]

Right branch passing through $(4, 6)$ and tending upwards as x

2^+ [B1]

(b) $x = 2$ **[B1]**

(c) $(0, -2)$ **[B2]**

If B0, M1 for either coordinate correct

(d) $(4, 6)$ **[B2]**

If B0, M1 for either coordinate correct

(e) Identifies the critical values -2 and 6 from the local extrema **[M1]**

$-2 < k < 6$ **[A1]**

[Back to Question] *[Total Marks: 10]*